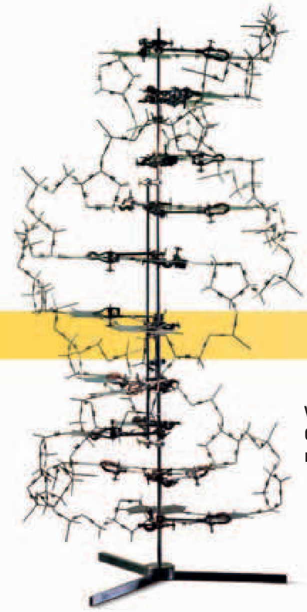
THE STORY OF GENETICS

THE SEARCH FOR THE BLUEPRINT OF LIFE AND THE MECHANISM OF HEREDITY



Watson and Crick's DNA model

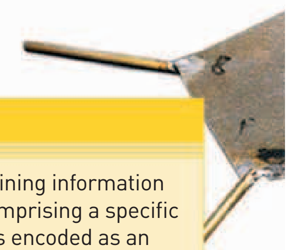
For thousands of years, humans have wondered how characteristics are inherited, but it was not until the 19th century that scientists began to understand the fundamental mechanisms. Now, the knowledge that DNA carries genetic information has provided insight into the basis of life itself.

One of the earliest theories of heredity was that of the ancient Greek Hippocrates, who proposed that elements from all of the body became concentrated in semen, which then made a human in the womb containing the characteristics of both parents. Charles Darwin later called this mechanism of inheritance "pangenesis."

It was not until the 19th century that the basic rules of heredity were discovered, by the Austrian monk Gregor Mendel. At about the same time, the Swiss scientist Friedrich Miescher extracted from the cell nucleus a substance he called "nuclein" (now known as DNA). In the early 20th century, American biologist Thomas Hunt Morgan's experiments with fruit flies confirmed that genes reside on chromosomes. However, it was still thought that protein, not DNA, was the substance that transmits inherited traits.

THE SIGNIFICANCE OF DNA

In the 1940s, Oswald Avery, Colin MacLeod, and Maclyn McCarty discovered that DNA is the hereditary molecule in most organisms and is the chemical basis of genetic information. In the early 1950s, Maurice Wilkins and Rosalind Franklin discovered that DNA has a helix shape, and in 1953, these findings were put together by Francis Crick and James Watson in their double-helix model of DNA. The Human Genome Project went further, mapping all the human genes.



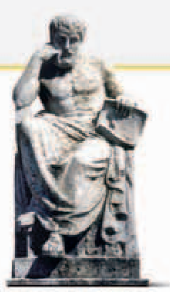
GENETIC CODE

A gene is a portion of DNA containing information for making a specific protein (comprising a specific sequence of amino acids). This is encoded as an "alphabet" of bases: A (standing for adenine), C (cytosine), G (guanine), and T (thymine). These are arranged into "words" (codons) of three letters; each codon corresponds to a particular amino acid. In a cell, a gene's codon sequence is translated into a sequence of specific amino acids, making the specific protein coded for by that gene.

> **"YOU,** YOUR JOYS AND YOUR SORROWS, YOUR MEMORIES AND AMBITIONS, YOUR SENSE OF PERSONAL IDENTITY AND FREE WILL, **ARE ... NO MORE THAN THE BEHAVIOUR OF... NERVE CELLS AND ... MOLECULES."**

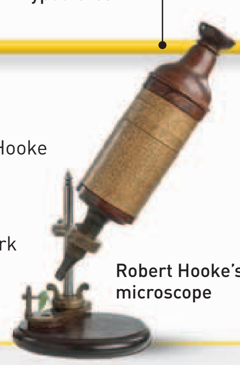
Francis Crick, *The Astonishing Hypothesis: The Scientific Search for the Soul,* 1994

c. 460–375 BCE
Hippocrates' pangenesis hypothesis
Hippocrates devises the theory that hereditary material collects from throughout the body and reassembles inside the womb to form human life.



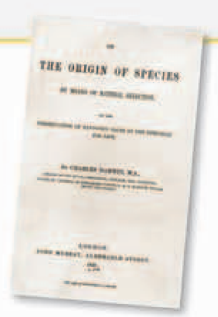
Hypocrates

1663–65
Cells first described
English scientist Robert Hooke coins the term "cell" to describe the microscopic units he observed while examining a section of cork with an early compound microscope.



Robert Hooke's microscope

1859
Theory of natural selection
Charles Darwin publishes *The Origin of Species,* in which he puts forward his theory that the fittest organisms survive and pass on their traits.



The Origin of Species

1863
Gregor Mendel
Experimenting with peas, Austrian monk Gregor Mendel finds that traits, such as whether peas are round or wrinkled, are passed on by independent units, later called genes.

Round and wrinkled peas

c. 1868–69
Nuclein discovered
Swiss scientist Friedrich Miescher discovers a substance he calls "nuclein" in the nuclei of white blood cells. Later called nucleic acid, nuclein is now known as DNA.

White blood cell

1880s
Meiosis discovered
Meiosis, the process of cell division that produces gametes (sex cells), is described in the early 1880s. Its significance for inheritance is elucidated in the 1890s by German biologist August Weismann.

1888
Chromosomes discovered
German anatomist Heinrich Waldeyer notices that the central part of the cell (the nucleus) sometimes contains threadlike bodies, for which he coins the term "chromosomes."